MILITARY SPECIFICATION

FUZE, MT, M563 ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Fuze, MT, M563 Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitations for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-60412	- Packaging, Packing and Marking for Shipment of Artillery Type and Rocket Fuzes, General Specification for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-331	- Fuze and Fuze Components Environmental and Performance Tests for
MIL-STD-1167	- Ammunition Data Card

FSC 1390

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2.1 (Cont'd)

DRAWINGS

Munitions Command

C8886458	- Box, Wirebound, Packing, Ammunition for Artillery Type and Rocket Fuzes (within M2A1 box.)
F8864492	- Box, Ammunition, Metal for Artillery Type and Rocket Fuzes
D10520688	- Fuze, MT, M563 Assembly
IEL-10520688	- Index of Inspection Equipment List for Fuze MT, M563, Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N.Y. 10018).

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced standards, and specifications, except as provided in 6.1.

3.2 Dynamic regulation.-The movement assembly shall be regulated in accordance with the requirements specified on the applicable drawing.

3.3 Environmental.

3.3.1 Jolt.-The fuze shall comply with MIL-STD-331, test No. 101.

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3.3.2 Jumble.-The fuze shall comply with MIL-STD-331, test No. 102.

3.3.3 Waterproofness.-The fuze shall comply with MIL-STD-331, test No. 108 with the exception that the duration of immersion shall be five (5) minutes and the pressure shall be three (3) psi guage.

3.3.4 Transportation-vibration.-The fuze shall comply with MIL-STD-331, test no. 104.

3.3.5 Temperature and humidity.-The fuze shall comply with MIL-STD-331, test No. 105.

3.4 Ballistics.

3.4.1 Muzzle action.-The fuze shall initiatate detonation of the round at 0 to 50 feet from the muzzle of the weapon within the temperature range of minus 45 degrees to plus 145 degrees Fahrenheit (F) when fired under the following conditions:

Weapon	Charge	Setting
105MM Howitzer w/M103 Tube	Zone 7	MA

3.4.2 Time-to-burst.-The fuze must initiate detonation of round during flight within the time limits specified in Table I, when fired under the following conditions:

Weapon	Charge	Setting
105MM Howitzer w/M103 Tube	Zone 7	½ seconds (sec.)
105MM Howitzer w/M103 Tube	Zone 7	5 sec.

3.4.3 Post transportation-vibration.-Subsequent to subjection to the transportation vibration test specified in MIL-STD-331, test No. 104, the fuze must initiate detonation of the round during flight when fired under the following conditions:

Weapon	Charge	Setting
105MM Howitzer w/M103 Tube	Zone 7	MA
105MM Howitzer w/M103 Tube	Zone 7	10 sec.

3.4.4 Excess pressure.-The fuze must initiate detonation of the round during flight when fired as follows:

Weapon	Charge	Setting
105MM Howitzer w/M103 Tube	112% of Zone 7 Pressure	MA
105MM Howitzer w/M103 Tube	112% of Zone 7 Pressure	10 sec.

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3.4.5 Low temperature.-The fuze must initiate detonation of the round during flight when the fuzed round and propellant are maintained at -40degrees - 5 degrees F, and fired as follows:

Weapon	Charge	Setting
105MM Howitzer W/M103 Tube	Zone 7	Muzzle action
105MM HOwitzer w/M103 Tube	Zone 7	10 sec.

3.4.6 High temperature.-The fuze must initiate detonation of the round during flight when the fuzed round and propellant are maintained at +140 degrees + 5 degrees F, and fired as follows:

Weapon	Charge	Setting
105MM Howitzer w/M103 Tube	Zone 7	Muzzle action
105MM Howitzer w/M103 Tube	Zone 7	10 sec.

3.4.7 Long Range.-The fuze shall be fired for information purposes at ambient temperature condition (at high and low temperature also as defined above during first article sample) as follows:

Weapon	Charge	Setting
105MM Howitzer w/M103 Tube	Zone 7	60 sec.

3.5 The fuze shall be shipped with (MA) Muzzle action line on lower cap in line with zero line on body.

3.6 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.6.1 Metal defects.-All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.6.2 Surface finish.-The requirements for surface finish are as detailed on the applicable drawings and ANSI-B46.1.

3.6.3 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawing.

3.6.4 Burr.-No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.5 Foreign matter.-No part or assembly shall contains chips, dirt, grease, rust, corrosion or other foreign matter. The cleaning methods and agents shall not adversely affect the finish nor contiminate the parts.

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3.6.6 Protective finish.-Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of contour configuration.

TABLE I

FUZE, MT, M563, ASSEMBLY

Weapon	Charge	Fuze Setting (sec)	Upper Limit (sec)	Lower Limit (sec)
105MM Howitzer w/M103 Tube	7	½	1.234	.153
105MM Howitzer w/M103 Tube	7	5	5.707	4.731

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Lot formation.-Each inspection lot of fuzes shall be comprised of components (except explosive components) each produced by one manufacturer by one process in accordance with the same drawing and revision and specifications and revisions, as applicable, and assembled on one assembly line. Each explosive component within a lot of assembled fuzes shall be limited to one manufacturer and one interfix number.

4.2 First article sample.

4.2.1 When required by the contract, the first units of production shall be submitted as an initial production sample. The sample shall consist of the following:

360 Component parts (as delineated in 4.2.2)

325 Subassemblies (as delineated in 4.2.3)

136 Fuze Assemblies (as delineated in 4.2.5 and 4.2.6)

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4.2.1 (Cont'd)

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and material, including packaging and packing shall be the same as used in regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.-After inspection by the contractor and provisional acceptance at source by the Government, 115 fuze assemblies shall be forwarded to a designated proving ground as specified in the contract. Three hundred and sixty (360) component parts, 325 subassemblies and 31 fuze assemblies shall be inspected, as provided herein, at a Government laboratory or such other facility as specified in the contract.

4.2.2 Component parts.

4.2.2.1 Twenty (20) of each component shall be tested as in 4.3.3.2 (Hardness), and 4.3.3.3 (Load).

4.2.2.2 Twenty (20) of each component shall be tested as in 4.3.3.4 (Bend).

4.2.2.3 Twenty (20) of each component shall be tested as in 4.3.3.5 (Compression).

4.2.3 Subassemblies.

4.2.3.1 Twenty-five (25) of each subassembly shall be tested as in 4.3.3.6.1 (Static-Load).

4.2.3.2 Twenty-five (25) of each subassembly shall be tested as in 4.3.3.6.2 (Torque).

4.2.4 Acceptance criteria.-Failure of any component or subassembly to comply with the requirements of the applicable drawing shall be cause for rejection of the initial production sample.

4.2.5 Fuze assemblies-Thirty-one (31) fuze assemblies shall be subjected to the following MIL-STD-test, with the fuze set on "Muzzle Action"(MA) position.

Test	Quantity	Requirement	Acceptance Criteria	Test Method
Jolt	4	3.3.1	4.2.5.1	4.4.24.1
Jumble	4	3.3.2	4.2.5.1	4.4.24.2
Waterproofness	4	3.3.3	4.2.5.1	4.4.24.3
Transportation-Vibration	5	3.3.4	4.2.5.1	4.4.24.4
Transportation-Vibration (Post)	10	3.3.4 & 3.4.3	4.2.6.1.e	4.4.24.4.1
Temperature Humidity	4	3.3.5	Information	4.4.24.5

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4.2.5.1 Acceptance criteria.-The criteria of the applicable MIL-Standard shall apply as determined by disassembly and examination in accordance with the procedures in 4.4.24. Where the accept/reject decision is based on engineering judgement, concurrence by the responsible Technical Agency must be obtained.

4.2.6 Ballistics.-One hundred and fifteen (115) of the initial production sample shall be subjected to the following ballistic test phases:

Phases	Quantity	Requirement	Test Method
Muzzle action	20	3.4.1	4.4.25.1
Time-to-Burst	40	3.4.2	4.4.25.2
Post-Transportation Vibration	10**	3.4.3	4.4.25.3
Excess Pressure	10	3.4.4	4.4.25.4
Low Temperature	10	3.4.5	4.4.25.5
High Temperature	10	3.4.6	4.4.25.5
Long Range	15	3.4.7	4.4.25.7

** Taken from the 15 fuzes subjected to the Transportation-Vibration Test.

4.2.6.1 Acceptance criteria.-The occurrence of any of the following shall be cause for rejection of the initial production sample.

a. More than one prior function when set for time (see 6.2.1).

b. Failure of more than one round to function within 50 feet in the muzzle action phase.

c. More than two time failures (see 6.2.2) or two outlier-offtimes (see 6.2.5) or one time failure and one outlier-offtime in the time-to-burst phase (see 3.4.2).

d. Failure to meet the time limits in Table I by an estimated lot percent defective greater than the appropriate value given in Table II for each time-to-burst phase (see paragraph 4.4.25.2).

e. Failure of more than one round to function within 50 feet when set for muzzle action or more than one time failure when set for time in any of the following phases: post transportation-vibration, excess pressure, low temperature or high temperature.

f. Long range phase shall be fired for information purposes only.

4.3 Inspection provisions.

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4.3.1 Submission of product.-The product shall be presented in accordance with MIL-STD-105.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a single defect basis in accordance with MIL-STD-105. For the classification of defects contained herein, the AQL given for each shall be used. All nonconforming material shall be rejected. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis for surface roughness determination. Four power (4X) magnification may be used in performing examination for visual defects.

4.3.2.1 Plate No. 1 First Lamina, Drawing F10523181 covering a detail of Drawing D10520688.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Diameter of holes M, A, B and 20, incorrect	Gage
102. Location of holes M, A, B and 20, incorrect	Gage
*103. M hole bridge bent beyond max. flatness requirement (to be checked within .040 min. diameter of hole)	Gage
104. Surface finish of holes M, and 20 improper	Visual
105. Chamfer on holes M and 20, missing or incorrect	Visual
106. Metal burred or damaged	Visual
107. Surface finish of thrust bearing surface of holes M and 20, improper	Visual

* Inspection to be performed on a 100% basis

4.3.2.2 Plate No. 2 Second Lamina, Drawing F8595968 covering a detail of Drawing D10520688.

CRITICAL

None defined

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4.3.2.2 (Cont'd)

Defect		Method of Inspection
MAJOR	AQL = 1.0%	
101.	Diameter of holes M, 1, 20, 21, 32 and 33 incorrect	Gage
102.	Location of holes M, 1, 20, 21, 32 and 33 incorrect	Gage
103.	Flatness incorrect	Gage
104.	Surface finish of holes M, 1, 20, 21, 32 and 33 improper	Visual
105.	Chamfer on holes M, 1, 20, 21, 32 and 33, missing or incorrect	Visual
106.	Metal burred or damaged	Visual
107.	Surface finish on thrust bearing areas of holes M, 1, 20, 21, 32 and 33 improper	Visual

4.3.2.3 Plate No. 4, F9205023 covering a detail of Drawing D10520688.

CRITICAL

None defined

MAJOR	AQL = 1.0%	
101.	Diameter of holes D, 17, 21, 33, P and N, incorrect	Gage
102.	Location of holes D, 17, 21, 33, P and N, incorrect	Gage
103.	Flatness, incorrect	Gage
104.	Chamfer on holes D, 17, 21, 33, P and N, incorrect	Visual
105.	Surface finish on thrust bearing areas of holes D, 17, 21, 33, improper	Visual
106.	Metal burred or damaged	Visual
107.	Surface finish of holes D, 17, 21, 33, improper	Visual

4.3.2.4 Plate No. 5, F9205024 covering a detail of Drawing D10520668.

CRITICAL

None defined

MAJOR	AQL = 1.0%	
101.	Flatness, incorrect	Gage
102.	Diameter of holes, 1, P and N, incorrect	Gage
103.	Location of holes 1 and P, incorrect	Gage
104.	Surface finish of hole #1, improper	Visual

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4.3.2.5 Plate No. 6, F11730349 covering a detail of drawing D10520688.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Diameter of holes 32, P and N, incorrect	Gage
102. Location of holes 32 and P, incorrect	Gage
103. Flatness, incorrect	Gage
104. Chamfer of hole 32, P and N, incorrect	Visual
105. Surface finish on thrust bearing areas of hole 32, improper	Visual
106. Metal burred or damaged	Visual
107. Surface finish of hole 32, improper	Visual

4.3.2.6 Escapement Lever and Arbor Assembly Drawing D10551798, covering a detail of Drawing D10520688.

CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Angularity of arbor flat with datum A and B incorrect	Gage
102. Angularity of pallets with datum C incorrect	Gage
103. Functional diameter of pallets incorrect (measured at gage line)	Gage
104. Metal burred, damaged	Visual
105. Surface finish improper	Visual

4.3.2.7 Housing, F10535664 covering a detail of Drawing D10520688.

CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. I.D. incorrect	Gage
102. O.D. of lip, incorrect	Gage
103. Eccentricity of I.D. with datum, excessive	Gage
104. Location of locating pin holes (3), incorrect	Gage
105. Location of adjusting stud hole, incorrect	Gage
106. Diameter of adjusting stud counterbore, incorrect	Gage
107. Location of hammer spring holes, incorrect	Gage
108. Surface finish of I.D., improper	Visual

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4.3.2.8 Arbor, Escapement, Drawing D10535526 covering a detail of Drawing D10520688.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Perpendicularity of lever seat with datum "C" incorrect	Gage
102. Diameter of pivots, incorrect	Gage
103. Length between bearing shoulders incorrect	Gage
104. Metal burred or damaged	Visual
105. Surface finishes on bearing surfaces, incorrect	Visual

4.3.2.9 Plate, Firing Pin Safety, Drawing C10521127 covering a detail of Drawing D10520688.

CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Surface finish improper (pits, dents, die marks, flats, etc. in the working surface contacting firing arm shaft to be checked after plating.)	Visual

4.3.2.10 Head, Drawing C10535672 covering a detail of Drawing D10520688.

CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Circular runout of forward diameter with datum "A"	Gage
102. Circular runout of rear diameter with datum "A"	Gage
103. Diameter incorrect at forward datum length	Gage
104. Diameter incorrect at rear datum length	Gage

4.3.2.11 Firing Pin, Drawing C10521154 covering a detail of Drawing D10520688.

CRITICAL	
None defined	

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4.3.2.11 (Cont'd)

Defect	Method of Inspection
MAJOR AQL = 1.0%	
101. Surface finish on shoulder angle, improper	Visual
102. Diameter of striker point below min.	Gage

4.3.2.12 Bushing Guide, C10533948 covering a detail of Drawing D10520688.

CRITICAL

None defined

MAJOR AQL = 1.0%	
101. Finish on bearing surfaces, improper	Visual
102. Position of locating surfaces, incorrect	Gage

4.3.2.13 Screw, Bottom Assembly - C10521153 covering a detail of Drawing D10520688.

CRITICAL

None defined

MAJOR 100% Inspection	
101. Length of screw shank, incorrect	Gage
102. Chamfer, incorrect	Visual

4.3.2.14 Gear, Escapement, Drawing D10542765 covering a detail of Drawing D10520688.

CRITICAL

None defined

MAJOR AQL = 1.0%	
101. Tooth profile, incorrect	Gage
102. Surface finish, improper	Visual

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4.3.2.15 Body, Drawing F10520690 covering a detail of Drawing D10520688.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Major diameter of external thread, improper	Gage
102. Pitch diameter of external thread, improper	Gage
103. Length to external seating shoulder, incorrect	Gage
104. External diameter incorrect at forward datum	Gage
105. Location of closing plug locating hole incorrect	Gage
106. External diameter incorrect at rear datum	Gage
107. Thickness through web of small threaded cavity, incorrect	Gage
108. Diameter of body assembly screw holes, incorrct	Gage
109. Location of body assembly screw holes, incorrect	Gage
110. Diameter of undercut above the external threads exceeds max.	Gage
111. Protective finish improper, inadequate or defective	Visual
112. Threads damaged or not full	Visual
113. Marking missing, incorrect or illegible	Visual
114. Location of movement assembly locating hole, incorrect	Gage

4.3.2.16 Cap, Lower-Drawing D10535661 covering a detail of Drawing D10520688.

CRITICAL	
1. Location of MA set line, incorrect	Gage
MAJOR AQL = 1.0%	
*101. Cumulative error of second scale exceeds max.	Gage
102. Diameter incorrect at forward basic length	Gage
103. Diameter incorrect at rear basic length	Gage
104. Eccentricity of forward diameter with datum A	Gage
105. Eccentricity of rear diameter at basic length datum A	Gage
106. Marking missing, incorrect or illegible	Visual
107. Protective finish improper, inadequate or defective	Visual

*To be checked from MA line, 5 sec., 10 sec., 15 sec., and 60 sec. graduation lines.

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4.3.2.17 Centrifugal Weights, D10542809, covering a detail of Drawing D10520688.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Overall length, incorrect	Gage
102. Diameter of hole in bent tab, incorrect	Gage
103. Location of hole in bent tab, incorrect	Gage
104. Flatness, incorrect	Gage
105. Parallelism, incorrect	Gage
106. Metal burred or damaged	Visual
107. Surface finish, improper	Visual

4.3.2.18 Firing Arm Assembly-Drawing C9212053 covering a detail of Drawing D10520688.

CRITICAL	
1. Fracture of upright bend of firing arm (prior to application of solid film)	Visual
2. Firing arm not staked to shaft	Visual
MAJOR AQL = 1.0%	
101. Insecure weight	Manual
102. Metal burred or damaged	Visual
103. Lubricant missing	Visual
104. Step or improper blend for safety plate drop off (Prior to application of solid film)	Visual
105. Thrust bearing surface damaged	Visual

4.3.2.19 Escapement Gear and Pinion Assembly Drawing C10542766, covering a detail of Drawing D10520688.

CRITICAL	
None defined	

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4.3.2.19 (Cont'd)

Defect	Method of Inspection
MAJOR AQL = 1.0%	
101. Concentricity of gear with datum	Gage
102. Perpendicularity of gear, incorrect	Gage
103. Metal burred or damaged	Visual

4.3.2.20 Plate No. 2, First Lamina Drawing F8595967, covering a detail of Drawing D10520688.

CRITICAL

None defined

MAJOR AQL = 1.0%	
101. Flatness incorrect	Gage
102. Diameter of holes #17, 65, 66, and D, incorrect	Gage
103. Location of holes #17, 65, 66, and D, incorrect	Gage
104. Surface finish on hole #17, incorrect	Visual
105. Chamfer on #17 hole, incorrect	Visual
106. Metal burred or damaged	Visual
107. Surface finish of thrust bearing area of hole #17, improper	Visual

4.3.2.21 Plate No. 7, Drawing F10551658, covering a detail of Drawing D10520688.

CRITICAL

None defined

MAJOR AQL = 1.0%	
101. Angular surfaces, incorrect	Gage
102. Location of holes D, T, 3, 92, 93, 150, 151, 152, and 153, incorrect	Gage
103. Diameter of holes D, R, T, 3, 92, 93, 150, 151, 152, and 153, incorrect	Gage
104. Surface finish on bearing surface of holes D and 3, improper	Visual
105. Surface finish of holes D, R, T, 3, 150, 151, 152, and 153, improper	Visual
106. Metal burred or damaged	Visual
107. Surface finish on angular surface, improper	Visual

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4.3.2.22 Closing Plug, Drawing 10535653, covering a detail of Drawing 10520688.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Location of locating pin hole, incorrect	Gage

4.3.2.23 Closing Plug Loading Assembly, Drawing C9205106, covering a detail of Drawing D10520688.

CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Angular flash hole not clear thru to relay	Visual
102. Relay inserted improperly	Visual
103. Crimped metal ends above plug surface	Visual
104. Metal burred or damaged	Visual
*105. Spinning or crimping of relay inadequate	Visual

*Inspection to be performed on 100% basis

4.3.2.24 Detonator Holder Assembly, Drawing C9205110, covering a detail of Drawing D10520688.

CRITICAL	
1. Detonator extends above flush	Visual
MAJOR AQL = 1.0%	
101. Detonator inserted incorrectly (brown end not facing cushion)	Visual
102. Extension of detent pivot pins exceed max. allowable	Gage
103. Crimping or staking inadequate	Visual
104. Crimping or staking extends above flush of plug surface	Visual
105. Metal burred or damaged	Visual

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4.3.2.25 Detonator Holder and Interrupter Assembly, Drawing C10533945 covering a detail of Drawing D10520688.

Defect	Method of Inspection
CRITICAL	
1. Interrupter springs disengaged from cover	Visual
MAJOR AQL = .40%	
101. Interrupter slides not free to open fully to the extent of exposing entire detonator face (check after tightening cover screws)	Manual/Visual
102. Detent pivot pins missing or loose	Manual/Visual
103. Insecure, improper or incomplete assembly	Manual/Visual

4.3.2.26 Body Assembly, Drawing C10520687, covering a detail of Drawing D10520688.

CRITICAL	
None defined	
MAJOR AQL = .40%	
101. Locating pin extension height, incorrect	Gage
102. Insecure, improper or incomplete assembly	Manual/Visual
103. Metal burred or damaged	Visual

4.3.2.27 Plate No. 7 Assembly, Drawing D10533947, covering a detail of Drawing D10520688.

CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Centrifugal weight hitting walls of slot for centrifugal spring	Manual/Visual
102. Heads of screws above flush of surface A	Gage
103. Raised metal on angular surfaces	Visual
104. Insecure assembly	Manual
105. Part missing	Visual
106. Metal burred or damaged	Visual

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4.3.2.28 Detonator Screw Assembly, Drawing C9205144, covering a detail of Drawing D10520688.

Defect	Method of Inspection
CRITICAL	
1. Detonator extends above flush of detonator screw	Visual
MAJOR AQL = 1.0%	
101. Overall length exceeds max. allowable	Gage
102. Green end of detonator not at bottom (thread end)	Visual
103. Crimping inadequate	Visual
104. Thread damaged	Visual

4.3.2.29 Escapement Lever Assembly, D10551145, covering a detail of Drawing D10520688.

CRITICAL	
None defined	
MAJOR AQL = .40%	
101. Perpendicularity of lever wings with datum A	Gage

4.3.2.30 Movement Subassembly, Drawing F10520691, covering a detail of Drawing D10520688.

CRITICAL	
1. Firing pin fired on "not fire" position of firing arm (Check prior to assembly of setback weights)	Gage
MAJOR AQL = .40%	
101. Firing pin does not fire on "must fire" position of firing pin	Gage
102. Firing pin protrusion less than min. in fired position	Gage
103. Burrs and irregularities which prevent proper seating of the movement in the fuze body	Visual
*104. Firing arm hits sides of plate #7 slots or mainspring surfaces during travel from fire to non-fire position (Check prior to assembly of timing disc support assembly)	Visual/Manual
105. Plates separated	Visual
106. Staking of screws inadequate	Visual

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4.3.2.30 (Cont'd)

Defect		Method of Inspection
107.	Foreign matter (within the mechanism)	Visual
108.	Metal burred or damaged	Visual
*109.	Protrusions above Plate No. 7 in path of centrifugal weights (Check prior to assembly of timing disc support assembly.)	Manual/Visual
*110.	Firing arm jammed by centrifugal weights or no. 7 plate (no clearance between firing arm weights and #7 plate which prevents backoff of firing arm - to be checked prior to assembly of timing disc support)	Manual/Visual

*Inspection to be performed on a 100% basis.

4.3.2.31 Movement Assembly, D10520689, covering a detail of D10520688.

CRITICAL

1.	Timing disc bushing loose on knurl of the arbor (to be examined prior to assembly of locknut)	Manual/Visual
2.	Timing disc retainer insecure, damaged or missing	Manual/Visual
3.	Escapement gear not locked sufficiently by lever pallet (to be checked prior to assembly to body)	Visual
4.	Timing disc slot touches firing arm upright when rotated	Manual/Visual

MAJOR AQL = .40%

101.	Firing pin striker point damaged	Visual
102.	Escapement lever not locked on proper side of pallet lock slide	Visual
103.	Improper or incomplete assembly	Visual
104.	Foreign matter	Visual
*105.	Inteference of timing disc support or timing disc retainer with firing arm in the cocked position	Manual/Visual
106.	Location of timing disc retainer	Gage
107.	Total height exceeds maximum	Gage
108.	Escapement spring broken, deformed or unseated	Visual
*109.	No firing arm upright play in the direction of arrow as shown (inspection to be performed prior to staking of timing disc retainer onto movement)	Manual
**110.	Protrustion of top of firing arm upright above timing disc less than minimum	Gage

*Inspection to be performed on a 100% basis.

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4.3.2.31 (Cont'd)

**Inspection to be performed on a 100% basis under the following conditions:

a. Arbor assembly pulled up with an applied load of 5 + 5 lbs so that upper bearing surface contacts the bottom of Plate #7.

b. Firing arm assembly held down so that the bottom bearing surface contacts the top of the laminated #2 plate.

4.3.2.32 Fuze Subassembly, Drawing F10520692, covering a detail of Drawing D10520688.

Defect	Method of Inspection
CRITICAL	
1. Escapement lever upright (s), missing, damaged or deformed	Visual
2. Hammerspring not engaging lug (safety) at 0.2 second setting	Manual/Visual
3. Adjusting stud not properly engaged in timing disc tab slot	Visual
4. Firing arm not in the unfired position	Visual
MAJOR AQL = .40%	
*101. Hammerspring not clearing lug (safety) at 0.5 second setting	Manual/Visual
102. Improper or incomplete assembly	Visual
103. Body assembly screws inadequately staked	Visual
*104. MA and zero line not in alignment prior to zero set operation	Visual
*105. Zero setting incorrect (after tightening adjusting stud)	Visual
*106. Adjusting stud insecure after tightening screw	Visual
*107. Timing disc tab not in contact with shoulder of adjusting stud	Visual
108. Setting torque below minimum	Gage

*Inspection to be performed on a 100% basis.

4.3.2.33 Fuze Assembly, D10520688, Less Detonator Holder and Interrupter Assembly, covering a detail of Drawing D10520688.

CRITICAL

None defined

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4.3.2.33 (Cont'd)

Defect	Method of Inspection
MAJOR AQL = .40%	
101. Detonator screw - above flush of body	Visual
102. Detonator screw assembly inserted improperly (Red end visible - improper)	Visual
103. Relay extends beyond flush	Visual
104. Relay improperly inserted in closing plug (Closed end not visible)	Visual
105. Staking of closing plug assembly screws inadequate	Visual
106. Foreign matter	Visual

4.3.2.34 Fuze Assembly, Drawing D10520688.

CRITICAL	
1. Interrupter in armed or partially armed position (Check immediately prior to assembly into fuze See 6.4.1 and 6.4.2)	Visual
MAJOR AQL = .40%	
101. Fuze not set on (MA) muzzle action. (MA line on lower cap not in line with zero line on body) to be checked immediately prior to packaging	Visual
102. Total length exceeds max.	Gage
103. Detonator Holder & Interrupter Assembly not flush or below surface A	Visual
104. Length to flange exceeds max.	Gage
105. Improper or incomplete assembly	Visual
106. Threads damaged	Visual
107. Sealant missing	Visual
108. Marking missing, incorrect or illegible	Visual
109. Dust seal missing or not coated	Visual
MAJOR 100% (Reference Drawing 10520692)	
*110. Adjusting stud not engaged in timing disc tab slot	Visual
*111. Timing disc tab not contacting adjusting stud shoulder	Visual
*112. Zero setting gap incorrect when lower cap is set on zero line	Gage

*To be checked through welsh plug holes after head, closing plug, detonator holder and interrupter assembly have been secured and markings applied.

(Rejected fuzes shall be resubmitted for a recheck of the tab engagement (Major 110 and 111) and zero setting gap (Major 112) after rework.)

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4.3.2.35 Packaging, packing, and marking.-The number of specimens and acceptance criteria for packaging, packing and marking shall be in accordance with MIL-P-60412.

4.3.2.36 Process-control audit.-Periodic unscheduled monitoring shall be conducted for the process operations on the following assemblies to determine compliance with the requirements of the applicable drawings. Bending adjustment of the firing arm upright is not permitted at any time.

a. Laminated Plates No. 1 and No. 2 Assembly Drawing F10542853
   (1) Staking adjusting blocks
   (2) Damaged or bent lever(Check prior to assembly of plates)

b. Movement Subassembly, Drawing 10520691
   (1) Application of lubricant
   (2) Staking of bottom assembly screws (Prior to regulation)
   (3) Mainspring pre-winding operation (3 turns)

c. Movement Assembly, Drawing D10542813
   (1) Regulation operation
   (2) Movement rewind after regulation
   (3) Backlash removal
   (4) Locknut torqued prior to timing disc torque test
   (5) Final lubrication

d. Fuze Subassembly Drawing F10520692
   (1) Tightening of hammerspring screws (after adjustment)
   (2) Application of lubricant to housing and hammerspring assembly
   (3) Lubrication and insertion of "O" ring between body and lower cap
   (4) Lock ring torque operation
   (5) Drilling operation for lockring setscrew
   (6) Tightening of lock ring setscrew
   (7) Zero setting procedure, including engagement of timing disc lug with adjusting stud

e. Fuze, MT, XM563E4 Assembly, Drawing D10520688
   (1) Insertion of dessicant pellet in closing plug loading assembly
   (2) Staking of assembly screws
   (3) Application of sealant and assembly torque of detonator interrupter assembly
   (4) Insertion of "O" ring between head and lower cap
   (5) Head torquing operation
   (6) Fuze setting torque
   (7) Application of sealant to screws and welsh plug

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4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Protective finish (Non-destructive).-The number of specimens and acceptance criteria for protective finish tests shall be in accordance with the applicable standards or specification listed on the drawing (See 4.4.1)

4.3.3.2 Hardness (Destructive).-Twenty samples from each lot of the following components shall be subjected to the hardness test. Where plating is required, the sample shall be unplated, where feasible. Failure of any component to comply with the requirements specified on the applicable drawing shall be cause for rejection of the lot (See 4.4.2).

Component	Drawing Number
Arbor, Escapement	D10535526
Gear, Escapement	D10542765
Pinion, Escapement	D10521101
Pinion, No. 1	D10521119
Pinion, No. 2	D10521120
Pinion, No. 3	D10521118
Pinion, No. 4	C10535698
Arbor	D11731758
Lever, Escapement	C10551071
Spring, Hammer	C10535667
Spring, Timing Disc	C9205083
Washer, Bearing	B10520292
Washer, Torque	C10520291
Firing Pin	C10521154
Washer, Timing Disc	B9205085
Collar	C9205040

4.3.3.3 Load (Destructive).-The same 20 timing disc springs that comply with the hardness test shall be subjected to the load specified on Drawing C9205083. Failure of any spring to comply with the requirement shall be cause for rejection of the lot (See 4.4.3).

4.3.3.4 Bend (Destructive).-Twenty firing arms from each lot shall be subject to the bend specified on Drawing C9205090. Any evidence of fracture of the firing arm, after bend, shall be cause for rejection of the lot (See 4.4.4).

4.3.3.5 Compression (Destructive).-Twenty firing pin springs from each lot shall be subjected to the compression specified on Drawing C10521132. Failure of any spring to comply with the requirement shall be cause for rejection of the lot (See 4.4.5).

4.3.3.6 Static load and torque (Destructive).

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4.3.3.6.1 Static load.-Twenty five (25) samples from each lot of the following subassemblies shall be subjected to the static load test. Failure of any subassembly to comply with the requirement specified on the applicable drawing shall be cause for rejection of the lot of subassemblies, as applicable (See 4.4.6).

Static Load	Drawing Number
Escapement Gear & Pinion Assembly	C10542766
Escapement Lever Assembly (Test Spring and Weights)	D10551145
Firing Arm Assembly (Test Pin & Arm)	D9212053
No. 1 Gear & Pinion Assembly	D10533955
No. 2 Gear & Pinion Assembly	C10533956
No. 3 Gear & Pinion Assembly	D10533957
No. 4 Gear & Pinion Assembly	C10535697
Pallet Lock Assembly (Pin)	B10521168
Mainspring Housing Assembly (Pins and Lug - Nondestructive) (Gear and Arbor Assembly - Destructive)	D10542885
Escapement Lever and Arbor Assembly	D10551798

4.3.3.6.2 Torque test.-Twenty-five (25) samples from each lot of the following subassemblies shall be subjected to the torque test. Failure of any subassembly to comply with the requirement specified on the applicable drawing shall be cause for rejection of the lot of subassemblies, as applicable (See 4.4.7).

Torque	Drawing Number
Escapement Lever and Arbor Assembly	D10551798
Mainspring Housing Assembly (Gear and Arbor Assembly - Destructive)	D10542885
Escapement Gear and Pinion Assembly	C10542766

4.3.3.7 Hammerspring deflection (Nondestructive).-The hammerspring of each housing and hammerspring assembly shall be subjected to the load specified on Drawing C10535663. Failure to comply with the requirements shall be cause for rejection of the assembly (See 4.4.8).

4.3.3.8 Dynamic regulation (Nondestructive).-Each (completely wound) movement shall be subjected to the regulation requirement of Drawing D10520689 for one (1) consecutive run of 100 seconds. Failure to comply with the requirement of drawing shall be cause for rejection of the assembly (See 4.4.9).

4.3.3.9 Operation - Movement Subassembly (Nondestructive) -Each movement subassembly shall be tested to determine compliance with the operation spin requirement on Drawing F10520691. Failure to comply with the spin requirement or movement to run the specified time shall be cause for rejection of the movement subassembly. (See 4.4.10).

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4.3.3.10 Kinetic energy of firing pin (Nondestructive).-Fifty movement subassemblies from each lot shall be tested to determine compliance with the requirement specified on Drawing F10520691. Failure of any subassembly to comply with the requirements shall be cause for rejection of the lot (See 4.4.11).

4.3.3.11 Static function test without oil on the escapement pallet and escapement gear teeth movement subassembly (Nondestructive).-Each movement subassembly produced shall be subjected to a static function test without the application of oil as specified on Drawing 10520691. Failure to comply with requirement shall be cause for rejection of the movement assembly (See 4.4.13).

4.3.3.12 Load-Setback weights (Nondestructive).-The setback weights within each movement subassembly shall be subjected to the load as specified on Drawing D10520689. Failure of any setback weight to comply with the requirement shall be cause for rejection of the movement assembly (See 4.4.12).

4.3.3.13 Spin function - Centrifugal weights - Movement Subassembly (Nondestructive).- The centrifugal weights within each movement subassembly shall be tested to determine compliance with the function requirements of Drawing F10520691. Failure to comply with the requirements shall be cause for rejection of the subassembly (See 4.4.14).

4.3.3.14 Spin non function - Centrifugal weights - Movement Subassembly (Nondestructive).-The centrifugal weights within each movement subassembly shall be tested to determine compliance with the nonfunction requirements of Drawing F10520691. Failure to comply with the requirements shall be cause for rejection of the subassembly (See 4.4.15).

4.3.3.15 Backlash - Movement assembly.-Each movement produced shall be tested to the torque requirement of Drawing 10520689 to determine that the movement is fully wound and the backlash is removed. Any visible motion of the timing disc when the specified torque is applied shall be cause for rejection of the movement assembly. (See 4.4.16).

4.3.3.16 Spin function - Pallet lock assembly - Movement assembly (Nondestructive).-The pallet lock assemblies within each movement assembly shall be tested to determine compliance with the function requirement of Drawing D10520689. Failure to comply with the requirement shall be cause for rejection of the assembly (See 4.4.17).

4.3.3.17 Spin nonfunction - Pallet lock assembly - Movement assembly.- The pallet lock assemblies within each movement assembly shall be tested to determine compliance with the nonfunction requirement of Drawing D10520689. Failure to comply with the requirements shall be cause for rejection of the assembly (See 4.4.18).

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4.3.3.18 Timing disc torque (Nondestructive).-The timing disc of each movement assembly shall be subjected to the torque specified on Drawing D10520689. Failure to comply with the requirement shall be cause for rejection of the movement assembly (See 4.4.19).

4.3.3.19 Arming - Detonator holder and interrupter assembly (Nondestructive).- Each safety interrupter shall be tested to determine the compliance with the arming requirement of Drawing C10533945. Failure to comply with the requirement shall be cause for rejection of the assembly (See 4.4.20).

4.3.3.20 Nonarming - Detonator holder and interrupter assembly (Nondestructive).-Each safety interrupter shall be tested to determine compliance with the nonarming requirement of Drawing C10533945. Failure to comply with the requirement shall be cause for rejection of the assembly (See 4.4.21).

4.3.3.21 Adjusting stud security body - Fuze Subassembly (Nondestructive).- After zero setting operation, each assembly produced shall be tested to determine security of the adjusting stud by application of the specified torque on Drawing F10520692. Any assembly that fails to comply shall be rejected. (See 4.4.22).

4.3.3.22 Setting torque fuze assembly (Nondestructive).-Each fuze assembly produced shall be tested to determine compliance with the setting torque requirement of Drawing D10520688. Failure to comply with the requirement shall be cause for rejection of the fuze assembly (See 4.4.23).

4.3.3.23 Environment tests (lots) (destructive).-Seventeen (17) fuze assemblies from each production lot shall be subjected to the following MIL-STD-331 tests, with the fuzes set on "Muzzle Action" (MA) (See 4.4.24).

Test	Quantity	Requirement	Acceptance Criteria	Test Method
Jolt	4	3.3.1	4.2.5.1	4.4.24.1
Jumble	4	3.3.2	4.2.5.1	4.4.24.2
Waterproofness	4	3.3.4	4.2.5.1	4.4.24.3
Transportation-Vibration	5	3.3.5	4.2.5.1	4.4.24.4

4.3.3.24 Ballistics (Destructive).-Seventy (70) fuzes from each production lot shall be subjected to the following ballistic test phases: (See 4.4.25).

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4.3.3.24 (Cont'd)

Phases	Quantity	Requirements	Test Method
Muzzle Action	20	3.4.1	4.4.25.1
Time-to-Burst	40	3.4.2	4.4.25.2
Longe Range	10	3.4.7	4.4.25.6

4.3.3.24.1 Acceptance criteria.-The occurrence of any of the following shall be cause for rejection of the lot the ballistic sample represents:

a. More than one prior function when set for time (See 6.2.1)

b. Failure of more than one round to function within 50 feet in the muzzle action phase.

c. More than two time failures (see 6.2.2) or two outlier-offtimer (see 6.2.5) or one time failure and one outlier-offtime in the time-to-burst phase (see 3.4.2).

d. Failure to meet the time limits in Table I by an estimated lot percent defective greater than the appropriate value given in Table II for each time-to-burst phase (See Paragraph 4.4.25.2).

TABLE II

Sample Size	Acceptable % Defective
20	4.09
19	4.14
18	4.17
17	4.22
16	4.26
15	4.31

e. The long range phase shall be fired for information purposes only.

4.3.4 Inspection equipment.-The examinations and tests shall be made using equipment listed in IEL-10520688. Equipment design shall be in accordance with the code designation on the applicable IEL.

4.4 Test method and procedures.

4.4.1 Protective finish.-The test procedures for protective finish shall be in accordance with the applicable specification referenced on the component drawing.

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4.4.2 Hardness.-Samples shall be tested in an area which is solidly supported in order to obtain valid readings. The median value of three hardness readings taken on each sample unit shall be used as the individual value.

4.4.3 Load.-The test shall be performed using the values as specified on Drawing C9205083. The timing disc springs that have been subjected to the dead-load test shall be scrapped.

4.4.4 Bend.-The firing arm shall be subjected to one twenty degree obtuse bend, min. The firing arms that have been subjected to the bend test shall be scrapped.

4.4.5 Compression.-The firing pin springs shall be compressed for the length of time specified on Drawing C10521132. The springs that have been subjected to the compression test shall be scrapped.

4.4.6 Static load.-All units shall be tested to the value specified on the applicable drawing. All units subjected to the static load test shall be scrapped.

4.4.7 Torque.-All units shall be tested to the value specified on the applicable drawing. All units subjected to the torque test shall be scrapped.

4.4.8 Hammerspring deflection (Housing and Hammerspring Assembly).-The required load shall be applied at the specified location parallel to the vertical axis of the assembly as specified on Drawing C10535663. The hammerspring must return to surface "C" when released.

4.4.9 Dynamic regulation monitoring.-The following procedure shall be observed: The movement assemblies shall be regulated, using the operational procedures furnished with the equipment. The movement assemblies shall be rotated in a clockwise direction when viewed from the bottom (No. 1 movement plate) at a constant speed of 15,000 $\pm$ 200 rpm. Timekeeping and performance shall be checked over a period of not less than 100 seconds. No adjustment shall be made on the escapement during the 100 second check run. Inspection shall be performed on a monitoring basis only (See 4.3.2.36.c(1).

4.4.10 Operation (Movement Subassembly).-The high spin function test shall be performed on a fully wound movement subassembly. The movement subassembly shall be rotated at the rpm specified in Drawing F10520691, in a clockwise direction when viewed from the bottom for a period of not less than 100 seconds at constant rpm to determine compliance with the requirement.

4.4.11 Kinetic energy of firing pin (Movement Subassembly).-The assembly shall be placed in a fixture and tested for kinetic energy by releasing the firing pin from the cocked position.

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4.4.12 Load-Setback weights.-The test shall be performed as outlined using the values as specified on Drawing D10520689.

4.4.13 Static function test without oil on the escapement pallet and escapement gear teeth (Movement Subassembly).-Simultaneously release both lever lock slides and determine that the movement starts and runs the minimum time required. A stop watch may be used to measure running time.

4.4.14 Spin function - Centrifugal weights (Movement Subassembly).-The spin function test shall be performed with the setback weights depressed or omitted. The subassemblies shall be rotated at the rpm specified on Drawing F10520691, to determine compliance with the requirement. The function test shall be performed prior to the nonfunction test (See 6.5.1).

4.4.15 Spin nonfunction - Centrifugal weights (Movement Subassembly).-The spin nonfunction test shall be performed. The subassemblies shall be rotated at the rpm specified on Drawing F10520691, to determine compliance with the requirement (neither centrifugal weight shall release) (See 6.5.2).

4.4.16 Backlash - Movement assembly.-The torque specified on Drawing 1 0520689 shall be applied to the lock nut in a clockwise direction. Care must be exercised in order that the maximum torque specified is not exceeded. This test shall be performed after the last wind.

4.4.17 Spin function - Pallet lock assembly (Movement Assembly).-The spin function test shall be performed by rotating the subassemblies, at the rpm specified on Drawing D10520689 to determine compliance with requirement. The function test shall be performed prior to the nonfunction test.

4.4.18 Spin nonfunction - Pallet lock assembly (Movement Assembly).-The spin nonfunction test shall be performed by rotating the assemblies, at the rpm specified on Drawing D10520689 to determine compliance with the requirement (Neither pallet lock assembly shall release). This test shall be the last spin test to which the movement is subjected.

4.4.19 Timing disc torque.-The torque test shall be performed after assembly of the timing disc assembly to the movement and assembly of the nut on the arbor. The highest and lowest reading will be taken in one uninterrupted complete revolution of the timing disc for determination of compliance with the requirements of Drawing D10520689.

4.4.20 Arming (Detonator holder and interrupter assembly).-The unarmed detonator holder and interrupter shall be rotated in a counter clockwise direction, when viewed from the interrupter cover. The interrupter slides must open to the extent of exposing the XM87 detonator when rotated at the rpm specified on Drawing C10533945.

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4.4.21 Nonarming (Detonator holder and interrupter assembly).-The unarmed detonator holder and interrupter shall be rotated in a counter clockwise direction, when viewed from the interrupter cover. The interrupter slides must not open when rotated at the rpm specified on Drawing C10533945.

4.4.22 Adjusting Stud Security - Fuze Subassembly.-The torque specified on Drawing F10520692 shall be applied in clockwise direction. Visible loosening shall be determined by the change in zero set gap.

4.4.23 Setting torque (Fuze Assembly).-The test shall be performed using the values as specified on the applicable drawing. The highest and lowest readings will be taken in one uninterrupted complete revolution of the lower cap for determination of compliance with the requirement of Drawing D10520688.

4.4.24 Environmental tests.-The procedures used shall be as set forth in the following sub-paragraphs, as applicable. Where spin testing is required, the non-arming or non-function test shall be performed prior to the arming test. The components of the disassembled fuzes shall be scrapped. All adverse conditions observed shall be reported to the responsible engineering element.

4.4.24.1 Jolt.-The test shall be performed in accordance with the procedure specifies in MIL-STD-331, Test No. 101 with the fuze orientations in the following sequence:

a. Nose down

b. Horizontal

c. Nose up

4.4.24.1.1 The fuze shall be disassembled and inspected as follows:

a. Prior to disassembly, check position of "MA" line relative to zero line. Check for security of the housing (Lower cap moves).

b. Remove head, (Holding lower cap securely).

c. Remove lower cap and check engagement of setting pin with timing disc lug, check centrifugal weights (both closed).

d. Remove detonator holder and interrupter assembly and check position of interrupter slides.

e. Remove closing plug and check relay.

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4.4.24.1.1 (Cont'd)

f. Remove movement from body. Check position of firing pin, firing arm upright, pallet lock slide and escapement lever upright. Check timing disc torque and for separation of movement plates.

g. Spin detonator holder and interrupter assembly for non-arming and arming.

h. Disassemble major sub-assemblies to the extent that all critical parts can be examined, such as gear and pinion assemblies, etc., for security of assembly, broken or bent pivots and teeth, fractured or bent firing arm or shaft or other part which might allow accidental release of firing pin. Check condition of all explosive elements.

4.4.24.2 Jumble.-The test shall be performed in accordance with the procedures specified in MIL-STD-331, Test No. 102 and inspected as outlined in 4.4.24.1.1.

4.4.24.3 Waterproofness.-The test shall be performed in accordance with the procedure specified in MIL-STD-331, Test No. 108, except as provided in 3.3.3. The fuze shall be tested with all vulnerable areas exposed. The fuze shall be disassembled and inspected as follows:

a. Remove head, detonator holder and interrupter assembly, lower cap, check all visible internal surfaces for evidence of water entry.

4.4.24.4 Transportation - Vibration .-The test shall be performed in accordance with the procedure specified in MIL-STD-331, Test No. 104, with the fuze orientations in the following sequence:

a. Parallel to longitudinal axis, nose up.

b. First transverse axis to be parallel to a line running through the marking "MA."

c. Second transverse axis to be parallel to a line running through the marking 23 (approximate).

4.4.24.4.1 Five (5) of the initial production sample of fifteen and five (5) of the production lot sample fuzes subjected to this test shall be disassembled and inspected as follows:

a. Prior to disassembly, check position of "MA" line relative to zero line.

b. Remove welsh plugs and check engagement of setting pin with timing disc lug.

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4.4.24.4.1 (Cont'd)

c. Check for security of the housing (Lower cap moves).

d. Remove head (holding lower cap securely).

e. Remove lower cap and check centrifugal weight (both closed).

f. Remove detonator holder and interrupter assembly and check position of interrupter slides. Spin assembly for non-function and function.

g. Remove closing plug and check relay.

h. Remove movement from body. Check position of pallet lock slide and escapement lever upright. Check timing disc torque. Spin movement for non-function and function.

i. Remove timing disc retainer support and check setback weights. Spin movement for centrifugal weight function.

j. Disassemble major sub-assemblies to the extent that all critical parts can be examined such as gear and pinion assemblies, etc., for broken or bent pivots and teeth, fractured or bent firing arm or shaft or other parts which might allow accidental release of firing pin. Check condition of all explosive elements.

4.4.24.4.2 Subsequent to the above safety determinations, the remaining ten (10) fuzes of the initial production sample shall be properly identified and fired as the post - transportation - vibration ballistic test sample denoted in 4.2.6. The sample shall be fired by the method set forth in 4.4.25.

NOTE: Discrepencies found in the above examination and tests which would definitely cause malfunction shall preclude testing of the ballistic portion of the initial production sample of units subjected to transportation - vibration.

4.4.24.5 Temperature - Humidity.-The test shall be performed in accordance with the procedure specified in MIL-STD-331, Test No. 105. The fuze shall be disassembled and inspected as follows:

a. Remove head, detonator holder and interrupter assembly, lower cap, housing assembly, movement assembly, and check all visible internal areas for excessive mositure and corrosion.

4.4.25 Ballistics.-The complete fuze assembly, Drawing D10520688 shall be assembled to a suitable projectile and fired as follows:

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4.4.25.1 Muzzle action phase.-Twenty (20) fuzes set with Muzzle Action (MA) line on "Lower Cap" in line with zero (0) line on "Body" shall be fired at ambient temperature in the weapon specified in 3.4.1.

4.4.25.2 Time-to-Burst Phase.-Forty (40) fuzes (20 at each required setting) shall be fired as specified in 3.4.2 under the conditions delineated in 4.3.3.24 as appropriate. The time to function of each fuze shall be measured to at least four decimal places by using two independent fuze chronograph systems. The time to function shall also be determined by using a Hawk Doppler velocimeter system. All measurements shall be recorded (See Paragraph 6.2.4).

4.4.25.2.1 Lot Percent Defective.-The time values obtained by the chronograph system shall be used to calculate the estimate of lot percent defective. When both chronograph systems provide a time value for an individual fuze, the average shall be taken as the individual fuze value. If one chronograph system fails to provide a time value, the one time value obtained from the functioning chronograph system shall be used as the individual fuze value. Individual fuze values shall be rounded off to the third decimal place using the method prescribed by the American Society for Testing Materials. The estimate of lot percent defective shall be calculated using the method prescribed in MIL-STD-414, Example B-3. For sample sizes 16 through 19, the estimated lot percent defective value shall be determined by interpolation between values given in Table B-5, MIL-STD-414 for sample sizes 15 and 20.

All appropriate time values obtained for each distance function phase shall be used in the initial calculation of sample average ($\bar{X}$) and standard deviation(s). The standard deviation value obtained shall then be used to determine statistical outliers (non-normal values) by using the criteria $\bar{X}$ plus or minus 2.88 times the standard deviation. A value judged to be an outlier shall be excluded and the remaining values used to recalculate $\bar{X}$ and S. A minimum of 15 statistically normal time values are required for each phase (See Paragraph 6.2.3).

4.4.25.3 Post Transportation - Vibration Phase.-Ten (10) fuzes - five (5) at muzzle action and five (5) at the 10 record setting - shall be fired for air burst at ambient temperature in the weapon specified in 3.4.3. Time-to-burst shall be recorded to the nearest millisecond for informational purposes, using two independent chronograph systems.

4.4.25.4 Excess Pressure Phase.-Ten (10) fuzes - five (5) at muzzle action and five (5) at the 10 record time setting - shall be fired for air burst at ambient temperature as specified in 3.4.4. Time-to-burst shall be recorded to the nearest millisecond for informational purposes, using two independent chronograph systems.

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4.4.25.5 Extreme Temperature Tests.-Twenty (20) fuzes shall be fired - five (5) at muzzle action setting at each extreme temperature specified in 3.4.6 and 3.4.7 and five (5) at the 10 second time setting at each extreme temperature specified in 3.4.6 and 3.4.7. Time-to-burst shall be recorded to the nearest millisecond for informational purpose, using two independent chronograph systems.

4.4.25.5.1 Temperature conditioning (-40°F and +140°F).-The test chamber shall be at ambient temperature at the start of the test and the chamber maintained at a temperature of minus 40°F or plus 140°F as applicable, until stabilization of the fuze temperature is reached (See 6.3). After stabilization, the fuze shall be saturated at minus 40°F or plus 140°F as applicable, for one hour, minimum.

4.4.25.6 Long Range.-Ten (10) fuzes shall be fired as specified in 3.4.7. The time to function of each fuze shall be measured for information purposes to the nearest millisecond by using two independent fuze chronograph systems.

4.4.25.7 Long Range Phase (for First Article Sample): During first article sample fifteen (15) fuzes shall be fired at the 60 second time setting at the following temperature condition (as specified in 3.4.7): 5 each at ambient temperature, 5 each at high temperature condition (140° + 5° F) and 5 each at low temperature condition (-40° - 5° F). Time to fuze function shall be recorded for information purposes to the nearest millisecond by using two independent fuze chronograph systems.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-Preservation and packaging shall be Level "A".

5.1.1 Level "A".-Packaging shall be in accordance with Drawing F8864492, except air pressure test is not required.

5.2 Packing.

5.2.1 Level "C".-Unit boxes shall be packed in accordance with Drawing C8886458.

5.3 Marking.-Ammunition boxes shall be marked in accordance with Drawing F8864492 for interplant shipment. Boxes shall be marked in accordance with Drawing C8886458. The ICC marking shall be "Percussion Fuzes - Handle Carefully."

5.4 Data cards.-Data card information shall be as specified in MIL-STD-1167.

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6. NOTES

6.1 Dimensions.

6.1.1 Compliance with tests (proving ground, ballistic performance and shop) required by the specifications and drawings is mandatory. The contractor will not assume, nor does the Government guarantee, that all possible combinations permitted by the tolerance limits of the specifications and drawings will consistently satisfy the test requirements. Therefore, the manufacturer is obligated to choose those combinations of tolerances and fits within the limits of the specifications and drawings that best suit his process needs and still satisfy the testing requirements.

6.1.2 Dimensions which are specifically referred to in the classification of defects, as listed in Section 4, are mandatory. Pertaining to other dimensions, it is recognized that current mass production methods may occasionally produce parts which are not in full compliance with all dimensional requirements, but which may provide adequate fuze performance. Therefore, the contractor, at his own risk, may utilize these parts provided there is no adverse effect on fuze functionability. This relaxation of requirements applies to "after the fact" deviations only, to dimensions only, and does not constitute authority to establish new dimensions or to relax efforts to achieve the required sizes within the specified limits. The procedure for accomplishing dimensional changes is available under the established engineering change system. Nothing above is to be construed as relieving the contractor of any responsibility in connection with performance of the fuze.

6.1.3 The contractor shall maintain records showing deviations from dimensional requirements, the extent of such deviations, the amount of material involved, and the date the material is introduced into the assembly line. Copies of these records shall be forwarded to the agency assigned custody of the technical data.

6.2 Ballistic parameters.

6.2.1 Prior Function.-Shall be defined as the occurrence of any of the following:

a. The round detonates in the bore of the gun except when fired MA.

b. The round detonates before 0.1 seconds when set at 0.5 seconds.

c. The round detonates before 3.5 seconds when set at 5 seconds.

6.2.2 Time failure.-A time failure is considered to have occurred if the round fails to detonate during flight.

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6.2.3 Where the number of statistically normal time values obtained for either time-to-burst phase is less than 15 and the lot is not rejected because of other reasons, the disposition of the lot shall be referred to the responsible procuring activity. The Government reserves the right to require the contractor to submit additional test samples. The number of additional test samples shall be sufficient to obtain a combined total of 20 values in order that the occurrence of outliers will not reduce the number of normal values obtained in the combined tests shall be used to determine lot acceptance or rejection. Attribute characteristic failures, e.g. duds occurring in the additional samples shall not be used to reject the lot.

6.2.4 Hawk-Doppler Velocimeter System.-Where only velocimeter system time measurements are obtained and these values indicate a premature or outlier which would affect lot disposition, the lot shall be referred to the responsible procuring activity. Hawk-Doppler values will not be used to determine sample average ($\bar{X}$) or standard deviation(s).

6.2.5 Outlier-offtime.-An outlier-offtime is a statistical outlier (non-normal value) which falls beyond the time limits specified in Table I.

6.3 Temperature stabilization of fuzes may be checked by a thermal measuring device in good thermal contact with the largest centrally located internal mass.

6.4 Armed/nonarmed interrupter.

6.4.1 Armed.-Both interrupter slides are out of position (face of detonator is exposed in the cover hole).

6.4.2 Partially armed.-One interrupter slide has remained in the outward position.

6.4.3 Nonarmed.-Both interrupter slides cover the hole in the plate.

6.5 Function/nonfunction centrifugal weights.

6.5.1 Function.-Both centrifugal weights have moved outward allowing the firing arm to enter between the weights.

6.5.2 Nonfunction.-Centrifugal weights remain intact and restrict the firing arm from moving in.

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6.6 Ordering Data.-Procurement documents should specify the following:

a. Title, number and date of this specification.

b. First article sample requirements (See 4.2).

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1390-A136

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-F-60655C(MU) Fuze, MT, M563 Assembly

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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